

Structural Cryptographic Engine (SCE)

Formal Security Model v1.0

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1 Introduction and Scope

This document defines a formal security model for the *Structural Cryptographic Engine* (SCE) as specified in the SCE v1.0 core specification and engine definition. The goal is to:

- Specify the syntax and semantics of SCE’s security-relevant components.
- Define adversary models and security goals.
- State clear assumptions under which SCE is intended to be secure.
- Provide a framework for cryptanalysis and future proofs.

This is a *model*: it does not constitute a proof that SCE is secure. Instead, it provides precise targets for analysis and proofs.

We focus on three levels:

1. The SCE Engine and SCE-DRBG (pseudorandomness).
2. The SCE AEAD mode (confidentiality and integrity).
3. Misuse resistance and structural security notions.

We intentionally do not reference any external frameworks or application-specific contexts; SCE is treated as a standalone cryptosystem.

2 Syntax of SCE Components

We describe SCE as a tuple of algorithms and stateful procedures.

2.1 Key and Parameter Space

Let:

- $\mathcal{K} = \{0, 1\}^{256}$ be the master key space.
- $\mathcal{N}$ be the nonce space (e.g. $\{0, 1\}^{96}$).
- $\mathcal{A} = \{0, 1\}^*$ be the associated data space.
- $\mathcal{P} = \{0, 1\}^*$ be the plaintext space.
- $\mathcal{C} = \{0, 1\}^*$ be the ciphertext space.

All strings are finite bit strings. We assume encodings for JSON or structured data as bit strings are fixed and unambiguous.

2.2 SCE Engine and DRBG Interfaces

The *canonical SCE Engine* is defined as a deterministic stateful algorithm:

$$\begin{aligned}\text{InitEngine} &: \{0, 1\}^{\ell_{\text{seed}}} \rightarrow S_0 \\ \text{StepEngine} &: S_t \rightarrow (S_{t+1}, M_t)\end{aligned}$$

where S_t is the internal state and $M_t \in \{0, 1\}^m$ is the mosaic output (after quantization).

The *SCE-DRBG* is a deterministic algorithm:

$$\begin{aligned}\text{SCE_DRBG.Init}(k_{\text{enc}}, \text{nonce}, AD, \text{SIV}) &\rightarrow \text{state}_{\text{drbg}} \\ \text{SCE_DRBG.Generate}(\text{state}_{\text{drbg}}, \ell) &\rightarrow (KS, \text{state}'_{\text{drbg}})\end{aligned}$$

where $KS \in \{0, 1\}^{8\ell}$ is a keystream of ℓ bytes.

2.3 SCE AEAD Interface

SCE AEAD is defined as a tuple of algorithms:

$$\begin{aligned}\text{SCE.Enc} &: \mathcal{K} \times \mathcal{N} \times \mathcal{A} \times \mathcal{P} \rightarrow \mathcal{C} \times \{0, 1\}^\lambda \times \{0, 1\}^\mu \\ \text{SCE.Dec} &: \mathcal{K} \times \mathcal{N} \times \mathcal{A} \times \mathcal{C} \times \{0, 1\}^\lambda \times \{0, 1\}^\mu \rightarrow \mathcal{P} \cup \{\perp\}\end{aligned}$$

with:

- $C \in \mathcal{C}$: ciphertext.
- $T \in \{0, 1\}^\lambda$: authentication tag.
- $\text{SIV} \in \{0, 1\}^\mu$: synthetic IV.

Semantically:

- $\text{SCE.Enc}(k, n, AD, P) = (C, T, \text{SIV})$.
- $\text{SCE.Dec}(k, n, AD, C, T, \text{SIV})$ returns P if and only if (C, T, SIV) is a valid SCE ciphertext under the inputs; otherwise it returns $\perp$.

3 Assumptions and Idealizations

We make standard cryptographic assumptions and one SCE-specific assumption.

Assumption 1 (Hash and MAC). *BLAKE2s, when used as a keyed MAC (BLAKE2s_k), behaves as a pseudorandom function (PRF) in its key for all practical purposes. When used unkeyed as a hash, it behaves as a collision-resistant hash function.*

Assumption 2 (HKDF). *HKDF with HMAC-SHA-256 behaves as a secure Key Derivation Function: its outputs are indistinguishable from random given the usual PRF assumptions on HMAC-SHA-256.*

Assumption 3 (SCE Engine Pseudorandomness). *For fixed parameters and key-derivation procedure, the combination:*

$$\text{EngineSeed} \mapsto (M_0, M_1, \dots)$$

where M_t are the mosaics extracted by the canonical SCE Engine, is indistinguishable from a sequence of independent random blocks to any efficient adversary not knowing the seed.

Remark 1. *Assumption 3 is the primary SCE-specific assumption. It is analogous to assuming that a DRBG's internal state update is pseudorandom. This assumption is what cryptanalysis must focus on.*

4 Security Goals for SCE-DRBG

We define pseudorandomness for SCE-DRBG via standard PRG games.

4.1 SCE-DRBG Pseudorandomness Game

Definition 1 (SCE-DRBG PRG Game). *Let $\mathcal{A}$ be a probabilistic polynomial-time (PPT) adversary with oracle access. In the real game, the challenger:*

1. *Samples a master key $k \leftarrow \mathcal{K}$.*
2. *For each adversary query (n, AD, SIV, ℓ) :*
 - (a) *Derives k_{enc} and the engine seed as specified.*
 - (b) *Initializes SCE-DRBG with (k_{enc}, n, AD, SIV) .*
 - (c) *Returns $KS \leftarrow \text{SCE_DRBG.Generate}(\cdot, \ell)$.*

In the ideal (random) game, step 2 is replaced by returning a uniformly random string of length ℓ bytes for each query, independent across queries.

The advantage of $\mathcal{A}$ is:

$$\text{Adv}_{\text{SCE-DRBG}}^{\text{prg}}(\mathcal{A}) = |\Pr[\mathcal{A} \text{ outputs } 1 \mid \text{real}] - \Pr[\mathcal{A} \text{ outputs } 1 \mid \text{ideal}]|.$$

SCE-DRBG is pseudorandom if this advantage is negligible in the security parameter.

5 Security Goals for SCE AEAD

We define confidentiality and integrity in standard AEAD terms, adapted to the presence of SIV and structured keystream.

5.1 Confidentiality: IND-CPA-like

Definition 2 (SCE AEAD IND-CPA Game). *Let $\mathcal{A}$ be a PPT adversary with access to an encryption oracle. The challenger:*

1. *Samples $k \leftarrow \mathcal{K}$.*
2. *$\mathcal{A}$ may query (n, AD, P_0, P_1) with $|P_0| = |P_1|$.*
3. *The challenger samples $b \leftarrow \{0, 1\}$ and returns $(C, T, SIV) \leftarrow \text{SCE.Enc}(k, n, AD, P_b)$.*
4. *$\mathcal{A}$ may repeat queries subject to standard AEAD restrictions.*
5. *Eventually, $\mathcal{A}$ outputs a guess b' .*

The advantage is:

$$\text{Adv}_{\text{SCE}}^{\text{ind-cpa}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

SCE achieves IND-CPA if this advantage is negligible.

5.2 Integrity: INT-CTXT

Definition 3 (SCE AEAD Ciphertext Integrity). *An adversary $\mathcal{A}$ interacts with an encryption oracle $\text{SCE.Enc}(k, \cdot)$ and eventually outputs a tuple $(n^*, AD^*, C^*, T^*, SIV^*)$. The adversary wins if:*

- *(C^*, T^*, SIV^*) was never returned by the encryption oracle on input (n^*, AD^*, P) for any P , and*
- *$\text{SCE.Dec}(k, n^*, AD^*, C^*, T^*, SIV^*) \neq \perp$.*

The advantage $\text{Adv}_{\text{SCE}}^{\text{int-ctxt}}(\mathcal{A})$ is the probability that $\mathcal{A}$ wins. SCE has ciphertext integrity if this advantage is negligible.

6 Misuse Resistance: Nonce Reuse and SIV

SCE uses a synthetic IV derived from the plaintext, associated data, nonce, and key. The aim is to reduce catastrophic failures under nonce reuse.

6.1 Nonce-Reuse Scenario

Consider an adversary that can force reuse of a nonce n across encryptions with different plaintexts and AD under the same key k .

Informally, we demand that:

- Reuse of n does not immediately expose a simple relation between plaintexts.
- The best the adversary can do is comparable to a random-DRBG AEAD with SIV (i.e. no trivial keystream reuse).

A more formal property can be expressed as:

Definition 4 (SCE-SIV Nonce-Misuse Resistance (Informal)). *Let $\mathcal{A}$ be allowed to choose (n, AD, P) and obtain (C, T, SIV) . The adversary may reuse n arbitrarily. SCE is nonce-misuse-resistant if, under Assumptions 1, 2, and 3, $\mathcal{A}$ cannot distinguish SCE ciphertexts from ciphertexts produced by an ideal SIV-based AEAD with random keystream, except with negligible advantage.*

7 Structural Security Goals

Beyond standard AEAD goals, SCE aspires to provide *structural* security properties.

7.1 Structure-Obfuscation Goal

Intuitively, we want:

- Given ciphertexts and tags alone, an adversary cannot infer high-level structural invariants of the plaintext beyond what is implied by length and trivial metadata.
- Even if SCE is used as a structural transform layer above AES, compromise of one layer does not trivially expose internal structural invariants.

7.2 State-Recovery Resistance

Another structural goal is:

- Given long transcripts of SCE-DRBG keystreams, an adversary cannot recover the engine's internal state S_t or its parameters without essentially inverting HKDF and BLAKE2s.

8 Summary of Security Assumptions

SCE v1.0's security claims rest on:

- BLAKE2s behaves as a PRF/MAC (Assumption 1).
- HKDF behaves as a secure KDF (Assumption 2).
- The canonical SCE Engine's mosaic sequence is pseudorandom and resists state recovery (Assumption 3).

Under these assumptions, the SCE-DRBG is intended to be a cryptographically secure PRG, and the SCE AEAD mode is intended to achieve:

- IND-CPA-like confidentiality,
- INT-CTXT-like integrity,
- SIV-style nonce-misuse resistance.

Additionally, SCE aims (non-formally) to provide:

- Structural obfuscation of model/data invariants,
- Resilience against internal attackers who see decrypted data but not structural decoding context.

9 Conclusion

This formal model clarifies:

- What SCE is supposed to protect.
- Which components are assumed secure.
- How SCE sits in standard AEAD and DRBG frameworks.

It also exposes where cryptanalysis should focus:

- The structural engine and mosaic extraction process.
- The interplay between engine dynamics and BLAKE2s compression.
- The behavior of SCE-DRBG under many seeds and parameter choices.
- Attempts to recover internal state or parameters from outputs.

SCE is not yet a proven scheme. However, this model provides a solid starting point for serious analysis and discussion.